CGIAR
BREEDING FOR
TOMORROW

Market Intelligence
Area of Work

MANUAL

IMPACT OPPORTUNITIES PORTAL & INVESTOR DASHBOARD



2025

Acknowledgment

We extend our heartfelt appreciation to all our partners who have contributed to the success of the Impact Opportunities Portal & Investor Dashboard. We are grateful for their expertise, commitment, and shared vision.

About CGIAR

CGIAR is a global research partnership for a food-secure future. CGIAR science is dedicated to transforming food, land and water systems in a climate crisis. Its research is carried out by 13 CGIAR Centers/Alliances in close collaboration with hundreds of partners, including national and regional research institutes, civil society organizations, academia, development organizations and the private sector.

www.cgiar.org

Report Contributors

Lenaerts, Bert, IRRI (b.lenaerts@cgiar.org)

Demont, Matty, IRRI (m.demont@cgiar.org)

GloMIP Team (glomip@cgiar.org)

Front cover and layout

Paguirigan, Neale Marvin, IRRI (n.paguirigan@cgiar.org)

Recommended citation

GloMIP. 2025. Impact Opportunities Portal & Investor Dashboard Reference Manual

1 Impact Opportunities Portal

Currently, no tools exist for comprehensively assessing impact opportunities across crops, countries and impact areas, much less CGIAR breeding portfolios. Therefore, the CGIAR Initiative on Market Intelligence has launched the Impact Opportunities Portal as part of the Global Market Intelligence Platform (GloMIP) (CGIAR, 2023) to facilitate evidence-based decision-making and priority setting at the crop level, utilising a live and user-centric dashboard.

The purpose of this section is to provide the end-user with a description of the system's functions, familiarity with the interface, and an understanding of how to navigate the Impact Opportunities portal correctly.

1.1 Navigation

A brief visual navigation guide is provided online. A longer descriptive walkthrough is provided below.

Users can customise the results within the Impact Opportunities Portal by changing any of the 16 entry fields (dropdown menu, input box, slider, push button) in the dashboard bar:

A. The **Indicator field** is linked to the Impact Area field and offers over 200 indicators organised by sub-Impact Area (see Table 3):
    - The Singe indicator subportal allows selecting only one indicator.
    - The Dual indicator subportal allows selecting two indicators.
    - The Multi indicator and Priority index subportals:
        - There is no limit on the number of selected indicators but exclude cross-cutting indicators from being selected.
        - By default, all indicators are selected. You need to deselect all before choosing an indicator group.
        - Selecting multiple options works additively: for example, selecting Future indicators and Breadth indicator will add all indicators labelled as either future or breadth to the final output. Multiple selections of the same indicator, that is, both individually and as part of an indicator group, will not lead to double-counting in the corresponding output.
    - Some indicators, like the number of supermarkets, are only available at the national data level, while others, such as the benefit indicators, are not available at the national level.
    - The symbol 🌐 indicates spatial data, and 📄 indicates tabular data measured at the national-crop level. Indicators without these symbols are measured at the national-level; therefore, national-crop-level values are allocated. 📈 indicates data is projected into the future. 💲 indicates benefit (rather than underdevelopment) indicators.
B. The **Data level field** enables the selection of datasets with differing smallest levels at which the data is generated:
    - The national scaling option offers crop-agnostic data at the country level.

- The national-crop scaling option offers data at the crop-country level using production (area, yield, price) and consumption (calorie supply) metrics from FAO for 50 crops.
- The market segment scaling option offers data at the market segment level using internally collected production metrics (area, yield, price) for 20 crops reflecting CGIAR priorities.
- For the national-crop and market segments scaling options: some indicators are measured directly at crop level or overlayed spatially with crop production; others are measured at country level but **downscaled** to crop level using either FAO or internal production and consumption metrics (see further).

C. The **Grouping field** offers different aggregation levels:
   - Indicators are **aggregated** from the measurement or allocation level (data level) to a higher grouping level (see further).
   - The national scaling option offers only country-based aggregation levels.
   - The national-crop scaling option also offers crop-based aggregation levels.
   - The market segment scaling option also offers breeding-based aggregation levels, such as breeding pipeline and trait.
D. The **Crop filter** allows a custom selection of crops to be included:
   - The national scaling option offers no crops (field disables).
   - The national-crop scaling option offers 50 crops.
   - The market segment scaling option offers 20 crops.
   - By default, all crops are selected. You need to deselect all, before choosing a crop group.
   - Selecting multiple options works additively: for example, selecting Legumes and Cereals will add all crops labelled as either legumes or cereals to the final output. Multiple selections of the same crop, that is, both individually and as part of a crop group, will not lead to double-counting in the corresponding output.
   - For details on crop names, see the see the Market Segment Portal Reference Manual.
E. The **Country filter** allows a custom selection of countries to be included:
   - The national and national-crop scaling options offer the same set of countries.
   - By default, all countries are selected. You need to deselect all, before choosing a country group.
   - Selecting multiple options works additively: for example, selecting Africa and Asia will add all countries labelled as either African or Asian to the final output. Multiple selection of the country, that is, both individually and as part of a country group will not lead to double-counting in the corresponding output.
F. The **Market Segment filter** allows a custom selection of market segments to be included (market segment scaling only).
G. The **weights** capture user preferences when combining different indicators into a **priority index** (Priority index subportal only, see further):
   - Weights across Impact Areas can be entered as any non-negative number, but are converted to proportions in the backend.

- Weights between breadth (magnitude) and prevalence or depth (intensity) indicators must be either 0 (only breadth), 25, 50, 75 or 100% (only prevalence and depth).
- Weights between current and future indicators must be either 0 (only current), 25, 50, 75 or 100% (only future). When only future indicators are preferred, the set of indicators is less balanced than when including also current indicators since there are **no future environmental and limited future equity indicators**.

H. The **Dynamic Graphs button** actives dynamic visualisations with labels but it takes more time to load.
I. The **Run button** needs to be pressed before new settings in the entry fields are activated.

1.2 Features

The Impact Opportunities portal features over 200 indicators across:

- the five CGIAR Impact Areas (Nutrition, Health and Food Security; Poverty Reduction, Livelihoods and Jobs; Gender Equality, Youth and Inclusion; Climate Change Adaptation and Mitigation; and Environmental Health and Biodiversity) and a cross-cutting category;
- three data levels (national, national-crop and market segment);
- 50 crops (including >20 CGIAR crops[1]);
- >170 countries (including +-100 with market segments);
- and >20 grouping levels (from region to trait).

Besides impact opportunity indicators, users can (1) select between national-level, national-crop-level and market segment-level data, choose (2) the aggregation or grouping level (region, country, crop, centre, etc.) and filter selected (3) crops (if relevant), (4) countries and (5) market segments (if relevant).

Impact opportunity indicators can be analysed separately (dubbed **single-indicator**), pairwise (**dual-indicator**), in group (**multi-indicator**) and through an index (**priority index**) incorporating investor preference weights across the Impact Areas, present and future indicators, and breadth and prevalence/depth indicators.

The Impact Opportunities Portal's output space consists of mainly **nonranked data visualisations** (treemaps, maps, scatter plots and area charts) and **data tables**. Visual representation of results was chosen to improve the accessibility of the dashboard to a mixed audience of breeders, social scientists, managers and donors. Visualisations are designed to be easy to read and interpretable without expert knowledge of statistics. Relatedly, figures are designed not to be suggestive—for example, visuals should not be ranked if ranking is not the main application of the data. Several functions are active in the backend to ensure smooth processing of the visual and non-visual outputs:

[1] Forages are a CGIAR crop but not yet included in the Impact Opportunities portal.

- Classify grouping level chosen into 3 or 5 groups of **equal length**, considering left-censoring, for colour-labelling;
- Plot labels containing the variable name, allocating mechanism and year;
- Rounding of numerical values using the following abbreviations: K (thousand), M (million), B (billion) and T (trillion).

An **Excel download** option is available for tabular data; images can be easily copy-pasted. Further, we provide meta-data and automated interpretation.

1.2.1 Crop allocation through statistical downscaling

Since most indicators are not measured directly at the crop level, we apply an **allocation mechanism** to estimate **crop-relevant** versions of the respective indicators.[2] For breadth indicators, we apply **statistical downscaling**[3], which assigns a proportion of the national-level value to each crop produced or consumed within the country (see Shirsath et al. (2020)).[4] For nutrition indicators, we use the proportion of either calories, proteins, iron, zinc and vitamin A (or a combination) consumed per crop, for poverty and equity indicators, we use the proportion of value created per crop (production value over GDP), and for environment and climate indicators, we use the proportion of area harvested per crop. Hence, nutrition indicators are downscaled using consumption levels, whereas poverty, equity, environment and climate indicators we use production levels. We calculate calorie consumption proportions for market segments by multiplying the national-crop level calorie consumption with the ratio of the (rural) population within the market segment over the whole population.[5] Statistical downscaling assumes a homogenous distribution of the indicator to be downscaled across the units to which we want to allocate the indicator. For example, we assume that dietary composition is similar for undernourished and non-undernourished people.

For a breadth indicator I, the downscaled values for a specific crop are calculated as **I•[%]National-Crop**, where [%]National-Crop represents the proportion of this crop in either the total amount of calories consumed, agricultural value produced or area harvested of the given country. For prevalence and depth indicators, we apply "pseudo-downscaling" by multiplying each indicator with the max-normalised downscales, [%]Country-Crop. Max-normalisation occurs by dividing by the maximum across the groups chosen. Multiplying with the normalised downscales assesses crop relevance and ensures that, like the breadth indicators, the allocated values for prevalence and depth indicators at the crop level are

---

[2] We indicate indicators that were measured directly at crop level (and hence, not allocated) using the symbols 🌐 (for spatial data) and 📄 (for tabular data).

[3] Downscaling is any procedure to infer high-resolution information from low-resolution variables, in this case to infer national-crop-level values from national-level data.

[4] We decided not to assign full national-level values to the country-crop profiles and market segments to avoid bias towards country-crop profiles and market segments in countries with larger production volumes. For example, the market segments identified in Brazil constitute five per cent of Brazil's total cropping area, whereas Indian market segments cover 30 per cent.

[5] Downscaling nutrition indicators at market segment level is challenging since market segments are based on production whereas hunger is driven by consumption.

always smaller or equal to the national-level values. This approach was inspired by the Biofortification Priority Index (Herrington et al., 2019) and yields comparable results (see Table 2A). The downscaling values used for allocating breadth indicators are calculated offline in advance, whereas normalising these downscales for allocating prevalence and depth indicators occurs live depending on the grouping and filters.

Some **spatial indicators**, such as those for poverty and the burden of hunger, are allocated using (1) **statistical downscaling** and (2) **rescaling** using modelled correlation with an **underlying spatial indicator** (the Relative Deprivation Index and infant mortality rate, respectively).

1.2.2 Aggregation

We use two mechanisms to aggregate indicators from the measurement or allocation level (national, national-crop or market segment level) to a higher grouping level. Summable or breadth indicators are summed[6]; we take the weighted average of nonsummable (prevalence and depth indicators) with weights chosen per indicator to consider size differences between countries. For example, when averaging the proportion of the national population living in poverty across India and Bangladesh, we use population as weights, giving higher importance to India.

1.2.3 Priority indexing (multi-indicator scoring)

The multi-indicator and priority index tabs enable comparing indicators irrespective of units used or Impact Areas chosen. We calculate the priority index— inspired by the Biofortification Priority Index (Herrington et al., 2019)—as the weighted average across normalised indicators through the following steps

- We **min-max-normalise (using winsorising for the 2% highest values)** each indicator across the grouping chosen, scaling the values between zero and one (not necessarily including zero).
- We extract 20 **first-principal-components** (by applying Principal Component Analysis) to avoid overlap in information between indicators (Filmer and Pritchett, 2001; Lecoutere et al., 2023). For example, poverty at $ 2.15/day is nested within poverty at $ 3.65/day; therefore, both indicators should not be considered two independent poverty indicators. The 20 groups are created by combining three dimensions: five Impact Areas, breadth versus prevalence/depth indicators and current versus future indicators. If selected, cross-cutting indicators are added to each Impact Area.
- We **weight** each of these 20 groups based on the user-specified weights[7] (correcting for non-applicable weights for groups without selected indicators).
- We calculate the **weighted sum** per Impact Area and across all Impact Areas.
- We **composition-normalise** the final six priority indices.

[6] The number of people undernourished in North Africa can be meaningfully obtained by summing the respective undernourishment values per country in North Africa.
[7] Impact Areas weights are not bounded since they are converted to composition percentages in the backend.

1.2.4 Benefit modelling

We capture nutritional benefits from **crop performance improvement** through the **reduced burden of chronic and hidden hunger** following the increased availability of macro- and micro-nutrients due to improved production (productivity) for producers. Further versions will also include improved production due to biofortification improvements for producers and (2) increased supply and lower consumer prices for consumers (Santeramo and Shabnam, 2015). We model the reduced burden of hunger following Lenaerts (2024). We model apparent relative nutrient intake levels by inversing the normalised relationship between undernourishment and relative calorie intake (Robinson et al., 2015), and applying the functional form to Disability-Adjusted Life Years (DALYs) lost due to nutritional deficiencies.

We quantify increased income due to improved economic surplus (increased production value for producers and reduced food costs for consumers) (Alston et al., 1995). Further versions will also include more or better-paying on-farm work for labourers, specifically vulnerable individuals engaged in agricultural and food value chains. Next, we quantified **reduced poverty** levels using poverty elasticities (Pauw and Thurlow, 2011; Thirtle et al., 2003).

Nutritional and poverty benefits are modelled as a **relative reduction in existing hunger and poverty levels**; we apply these reductions to the respective underdevelopment indicators to obtain the absolute reduction in the burden of chronic and hidden hunger and the number of poor people at $1.90/day. We capture crop performance in five dimensions: (1) productivity, (2) price premiums, (3) iron content, (4) zinc content and (5) vitamin A content (based on β-carotene content). We define **crop performance improvements** relatively based on expert estimates for two **scenarios** regarding breeding efforts and resources: conservative and optimistic.

1.3 Applications and interpretations

By summarising impact opportunities, the portal enables identifying hotspots for specific development challenges. **Surveying and hotspotting indicators** across countries, regions, crops, market segments and breeding pipelines can be used to:

- set priorities for **resource allocation**, such as within CGIAR breeding programs;
- request and justify **resource mobilisation** for proposal writing;
- check the current state of development across the five Impact Areas for inclusion in publications (reports, papers, book chapters, presentations);
- check whether market segmentation aligns with global needs and identify missing market segments;
- aid market segmentation (national cropping area, yield and price data from FAO form a starting point for market segmentation).

The **national(-crop)** data levels provide a **global** overview. For a more short-term view, users can focus on countries where **market segments** already exist. These countries are already **CGIAR priorities** and can thus be impacted more easily. Next, market segments served by a **breeding pipeline** constitute the quickest pathways to impact since **technology** is already underway.

Breadth indicators are the **most common** dimension to report impacts due to their ease of communication and **global scope** (breadth indicators can be summed easily). Alternatively, **prevalence indicators** provide a more **localised** view of impacts. For example, certain crops are only grown in limited climate zones or by indigenous people, but they might provide a majority of food for the people affected. Under such circumstances, breadth indicators may fail to capture significant opportunities to impact livelihoods on a local scale positively. Therefore, prevalence indicators can be used to check where investments in agriculture might have a significant effect locally—**targeting the most marginalised**—but not necessarily globally. The objective of using prevalence indicators is mitigating any potential bias in favour of activities yielding large outreach numbers yet exhibiting a relatively modest impact per individual or area (CGIAR, 2021b). Prevalence indicators ensure that transformative endeavours, albeit involving smaller populations or areas, receive equitable consideration. Lastly, depth indicators such as the poverty gap are ambiguous impact measures since they only measure the depth of the problem for those affected. For example, when people close to the poverty line are lifted out of poverty, the remaining people in poverty become poorer on average. Based on the poverty gap alone, this would reflect an increase in poverty—hence, a decrease in development, although people have been lifted out of poverty—clearly, a positive development outcome (Ravallion, 1992). Depth measures like the poverty gap can be meaningfully used with binary or scoped breadth indicators like the number of people in poverty. Similarly, modelled benefits can be subdivided into breadth, prevalence and depth indicators.

The Impact Opportunity indicators capture the **human/people dimension** by projecting the Impact Areas onto the respective population affected. Therefore, we survey where and how many people live in poverty, suffer from undernutrition, are exposed to climate hazards, where women, youth and indigenous people live, and where women suffer from gender inequality.

Users can select their preferred indicator in the Single-indicator subportal or combine multiple indicators through the Multi-indicator and Priority index subportals, enabling a higher-level view of impact opportunities. **Combining multiple indicators** improves the **robustness** of the corresponding breakdown across countries, regions, crops, market segments and breeding.

Since **downscaled** values are not measured at crop level, merely allocated, they represent **synthetic data** and hence, care should be taken when reporting or interpreting these results. The Impact Opportunities Portal should not be used to quote absolute values for the downscaled indicators but rather be used relatively, that is, compared to other values within the same grouping (for example, comparing poverty for maize across countries). Since scaled-down values are **directionally correct** rather than absolute estimates, we suggest using the Impact Opportunities Portal to provide rankings or relative distributions of groups across the grouping chosen for the selected indicator(s), countries, crops and market segments. Downscaling improves national-level indicators by **overlapping with crop production and consumption** to determine whether a **crop** has an **entry point** for impacting respective indicators. For example, countries with high diabetes rates but low rice consumption levels (regarding calories) provide limited opportunities for (low-glycaemic) rice to reduce the burden of non-communicable diseases.

2 Terminology

We distinguish between (1) (**under)development indicators**, which only consider market parameters (and hence, no technology parameters) and quantify underdevelopment levels, and (2) **(ex-ante modelled) benefit indicators**, which quantify technology-driven ex-ante impacts or underdevelopment alleviated assuming full adoption and no risks (see Table 1 for definitions) By assessing future changes, ex-ante modelled benefits can be further improved into **projected** benefits (considering existing patterns and trends) and **forecasted** benefits (considering historical data and anticipated future shocks). Most impact assessments consider absolute impacts in terms of people reached, such as the number of women targeted and the number of people lifted out of poverty—called **breadth indicators**. We further consider **prevalence and depth indicators**, representing relative values, irrespective of a target's size, such as the proportion of the population in poverty, greenhouse gas emissions emitted per unit of area harvested and the poverty gap. In sum, breadth refers to magnitude, prevalence to frequency and depth to intensity (CGIAR, 2021b; Ravallion, 1992; Tenny and Hoffman, 2023). Prevalence is determined at a higher level (for example, at the country level), whereas depth is measured at the household or individual level.

Underdevelopment and modelled benefit indicators can be further classified according to their information density (Table 2). Common indicators such as the number of people in poverty or undernourished provide limited information on the challenges of alleviating poverty and undernourishment since people are **binary** coded as either poor or non-poor depending on a cutoff, excluding the depth of poverty for each poor person (similarly for undernourishment). For other impact areas, binary indicators are uncommon due to the difficulty in identifying meaningful cutoffs—for example, greenhouse gas emissions. Continuous versions of these indicators quantify the underlying condition—the poverty gap for poverty and a measure of the burden of chronic hunger for undernourishment. Due to their increased complexity, **continuous** indicators are more difficult to communicate. The easiest indicators to communicate are **scoped** indicators capturing the target group.

Table 1. Overview of impact opportunity variables.

Category	Type	Definition	Summable	Example
Underdevelopment indicators	**Breadth**	Absolute indicator quantifying the magnitude of underdevelopment.	Yes	Population in poverty
	Prevalence	Relative indicator quantifying the frequency of underdevelopment, obtained by dividing the breadth indicator by the relevant overall population or area.	No	Proportion of the population in poverty (%)
	Depth	Relative indicator quantifying the intensity of underdevelopment per unit targeted.	No	Poverty gap
Helper variables	**Downscaling variable**	Composition (%) per country across crops of calories consumed, value produced or area harvested.	No	Proportion of calories consumed through rice in India
	Relative breadth indicator	Breadth indicator prorated by dividing by the area harvested or production volume, as reported by the corresponding data source.	No	Emissions per hectare
	Weighting variable	Absolute indicator quantifying the size of a certain crop, country or market segment.	Yes	Population
Benefit indicators	**Breadth***	Absolute indicator quantifying the magnitude of ex-ante assessed impacts (assuming full adoption and no risks) in absolute numbers reached.	Yes	People lifted out of poverty
	Prevalence	Relative indicator quantifying the frequency of ex-ante assessed impacts (assuming full adoption and no risks), obtained by dividing the breadth indicator by the relevant overall or affected population or area.	No	Proportion of the (poor) population lifted out of poverty (%)
	Depth**	Relative indicator quantifying the intensity of ex-ante assessed impacts (assuming full adoption and no risks) per unit affected.	No	Benefit per person relative to annual income
Cost indicators	**Variable cost**	Operational costs linked to maintaining and running breeding operations.	Yes	Labour cost of breeding team
	Fixed costs	Cost of setting up breeding operations.	Yes	Construction cost of greenhouse

Note: * Technology breadth, sometimes called reach, refers to the extent of a technology's effect. ** Technological depth captures technologies' expected impactfulness.

Table 2. Overview and examples of impact opportunity indicator types according to their information density.

		Continuous	Binary	Scoped
Breadth	1	Summed poverty gap*	Poor people	People targeted
	2	Summed DALYS due to calorie deficiency	Undernourished people	Same
	3	Summed DALYS due to hidden hunger	People suffering from hidden hunger*	Same
	4	CO_2e emissions	NAP	Area targeted
Prevalence	1	Average poverty gap/FGT indices	Proportion of poor people	People targeted
	2	Summed DALYS per capita due to calorie deficiency	Proportion of undernourished people	Same
	3	Summed DALYS per capita due to hidden hunger	Proportion of people suffering from hidden hunger	Same
	4	CO_2e emissions/ha	NAP	Area targeted
Depth	1	Mean depth of poverty as percentage of poverty line*		
	2	Mean depth of undernourishment as percentage of recommended calorie intake*		
	3	Mean depth of hidden hunger as percentage of recommended micronutrient intake*		
	4	NAP		

Notes: NAP: not applicable. FGT: Foster–Greer–Thorbecke. *refers to indicators that are not commonly reported.

3 Data

This section documents the data sources used in the Impact Opportunities Portal and Investor Dashboard. See Appendix B for details on the (offline) data preprocessing.

Most indicators are collected from various public data platforms like the World Bank and different UN agencies (FAO, WHO, ILO, UN Women, UN DESA). Selected datasets were obtained through personal communication, specifically climate hazards and greenhouse gas emissions. All data sources are listed in the **Indicator Catalogue** in the Impact Opportunities Portal. The main data criteria for inclusion were global coverage and data availability at the national or national-crop level. Therefore, more detailed data only available for selected countries, such as gendered market intelligence collected by the G+ tools, have not been included. For more information on data on market segments and Target Product Profiles (TPPs), see the Market Segment Portal and TPP Portal Reference Manuals. Only countries that are listed on FAOSTAT's "Crops and livestock products" subportal (FAOSTAT, 2022) are included.

The UN regional and subregional classification[8] follows UNSD (2023); the CGIAR regional classification[9] follows CGIAR (2021) (see Figure 1); the MS (that is, market segment) subregional classification was defined internally (see Figure 2) (see Standards and Manuals under Resources); focus countries are those countries with > 5% of the population poor at $6.85/day (2017 PPP); the sub-Saharan African countries classification follows the World Bank (2018); the food system classification follows Marshall et al. (2021); CGIAR crops are those crops for which market segments exists, non-CGIAR crops are globally important crops regarding area harvested, calories consumed and production value generated that are not included in the list of CGIAR crops (see Standards and Manuals under Resources); crop groups (including food security crops and diversity crops) were defined internally (see Standards and Manuals under Resources).

[8] Non-classified countries were classified based on best judgement.
[9] The same as above.

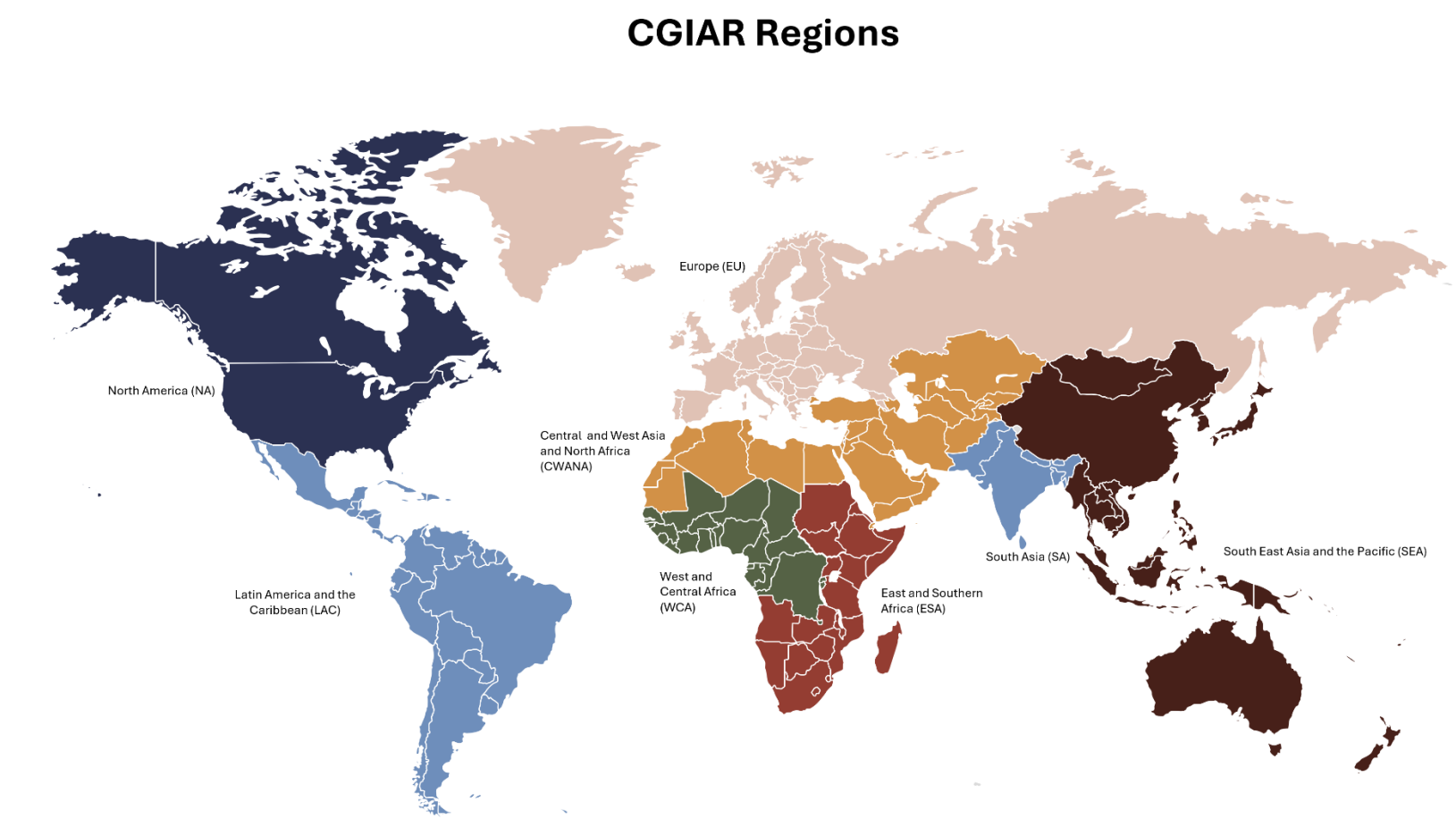
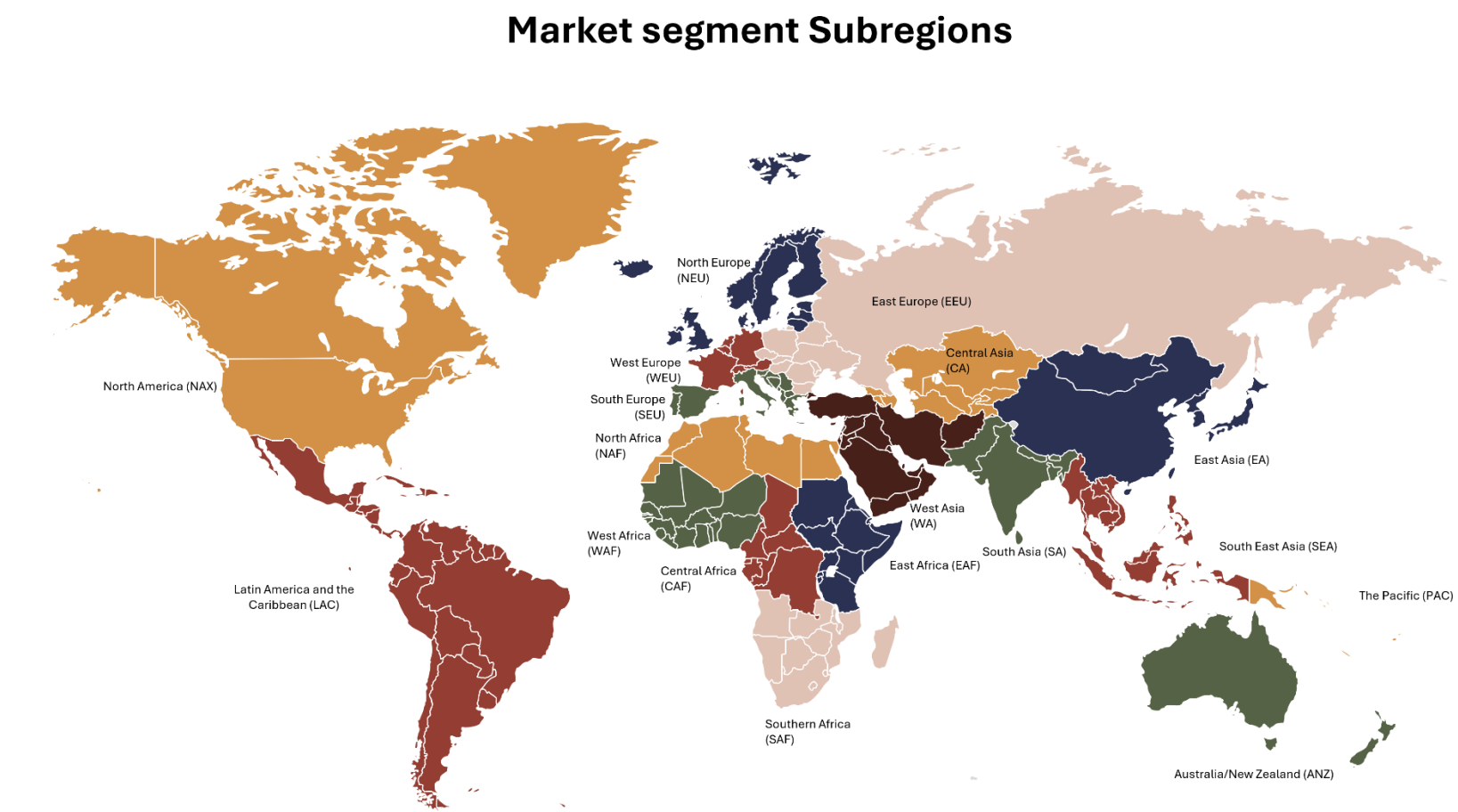


Figure 1 CGIAR Regions



Source: Breeding for Tomorrow Science Program

Figure 2 Market Segment Subregions

The Impact Opportunities Portal features indicators reflecting **underdevelopment related to agriculture and food** according to the five CGIAR Impact Areas and one cross-cutting category, which covers general descriptors of the agricultural sector, food consumption patterns, wider economy and value chain; each Impact Area is further broken down into sub-areas (Table 3).

Table 3. Sub-impact areas per CGIAR Impact Area

Nutrition, health and food security
- Nutrition crop improvement impact
- Macronutrient deficiency
- Micronutrient deficiency
- Non-communicable diseases (NCDs)
- Food utilisation
- Diet affordability
- Self-sufficiency
- Demographics

Poverty reduction, livelihoods and jobs
- Poverty crop improvement impact
- Poverty
- Rural employment
- Rurality
- Urbanisation
- Food production

Gender equality, youth and inclusion
- Gender-poverty
- Gender-hunger
- Youth-gender-hunger
- Gender-health
- Gender-production
- Gender-agency
- Gender-education
- Gender-overall
- Gender-demographics
- Youth-demographics
- Youth-gender-demographics
- Indigenous

Environmental health and biodiversity
- Natural resources*
- Environmental degradation
- Field inputs
- Biodiversity

Climate adaptation and mitigation
- Climate hazards
- GHG emissions

Cross-cutting
- Agronomy
- Economy
- Vendor
- Other

Note: *Surface freshwater is seen as a resource (whereas drought is based on rainfall and thus a climate variable).

4 Software

All steps are conducted in R (R Core Team, 2020): data cleaning and visualisation were performed using the package *tidyverse* (Wickham, 2021), and the application is built through the package *shiny* (Chang et al., 2023). The application is hosted online through shinyapps.io.

5 References

Alston, J. M., Norton, G. W., and Pardey, P. G. 1995. *Science Under Scarcity: Principles and Practice for Agricultural Research Evaluation and Priority Setting*, Ithaca, NY, Cornell University Press

CGIAR. 2021a. *One CGIAR Region Mapping*, https://docs.google.com/spreadsheets/d/1O5BtrGjuU49TFGb5xGTAr_joTQAHFuj9/edit?usp=sharing&ouid=113606872214300277418&rtpof=true&sd=true

CGIAR. 2021b. *Estimating Projected Benefits: Guidance for IDTs*, CGIAR

CGIAR. 2023. *Global Market Intelligence Platform.* <*https://glomip.cgiar.org*>

Chang, W., Cheng, J., Allaire, J., Sievert, C., Schloerke, B., Xie, Y., Allen, J., McPherson, J., Dipert, A., and Borges, B. 2023. shiny: Web Application Framework for R

Crop Trust. 2024. *Genesys*, https://www.genesys-pgr.org/ (date last accessed 19 February 2024)

FAOSTAT. 2022. <*http://www.fao.org/faostat/en/#data*>

Filmer, D. and Pritchett, L. H. 2001. Estimating wealth effects without expenditure data—or tears: an application to educational enrollments in states of India, *Demography*, vol. 38, no. 1, 115–32

Herrington, C., Lividini, K., Angel, M. D., and Birol, E. 2019. Prioritizing countries for biofortification interventions: Biofortification priority index second edition (BPI 2.0), *HarvestPlus Working Paper*, no. 40, Advance Access published 2019

Lecoutere, E., Mishra, A., Singaraju, N., Koo, J., Azzarri, C., Chanana, N., Nico, G., and Puskur, R. 2023. Where women in agri-food systems are at highest climate risk: a methodology for mapping climate–agriculture–gender inequality hotspots, *Frontiers in Sustainable Food Systems*, vol. 7

Lenaerts, B. 2024. *Quantifying the health impact of crop breeding: Revisiting the Disability-Adjusted Life Years Approach*, MedRxiv preprint

Marshall, Q., Fanzo, J., Barrett, C. B., Jones, A. D., Herforth, A., and McLaren, R. 2021. Building a global food systems typology: A new tool for reducing complexity in food systems analysis, *Frontiers in sustainable food systems*, 432

Pauw, K. and Thurlow, J. 2011. Agricultural growth, poverty, and nutrition in Tanzania, *Food policy*, vol. 36, no. 6, 795–804

R Core Team. 2020. R: A language and environment for statistical computing. R Foundation for Statistical Computing

Ravallion, M. 1992. *Poverty comparisons: A guide to concepts and methods*, The World Bank

Robinson, S., Mason-D'Croz, D., Islam, S., Sulser, T., Robertson, R., Zhu, T., Gueneau, A., Pitois, G., and Rosegrant, M. W. 2015. *The international model for policy analysis of agricultural commodities and trade (IMPACT): model description for version 3*, IFPRI Discussion Paper 01483, Washington, DC, IFPRI

Santeramo, F. G. and Shabnam, N. 2015. The income-elasticity of calories, macro-and micro-nutrients: What is the literature telling us?, *Food Research International*, vol. 76, 932–37

Shirsath, P. B., Sehgal, V. K., and Aggarwal, P. K. 2020. Downscaling regional crop yields to local scale using remote sensing, *Agriculture*, vol. 10, no. 3, 58

Tang, F. H., Nguyen, T. H., Conchedda, G., Casse, L., Tubiello, F. N., and Maggi, F. 2024. CROPGRIDS: a global geo-referenced dataset of 173 crops, *Scientific Data*, vol. 11, no. 1, 413

Tenny, S. and Hoffman, MR. 2023. *Prevalence*, Treasure Island (FL), StatPearls

Thirtle, C., Lin, L., and Piesse, J. 2003. The impact of research-led agricultural productivity growth on poverty reduction in Africa, Asia and Latin America, *World Development*, vol. 31, no. 12, 1959–75

UNSD. 2023. *Standard country or area codes for statistical use (M49)*, https://unstats.un.org/unsd/methodology/m49/

Wickham, H. 2021. tidyverse: Easily Install and Load the 'Tidyverse'. R package version 1.3.0

World Bank. 2018. *Atlas of Sustainable Development Goals 2018: World Development Indicators*, Washington, DC, World Bank

Appendix A

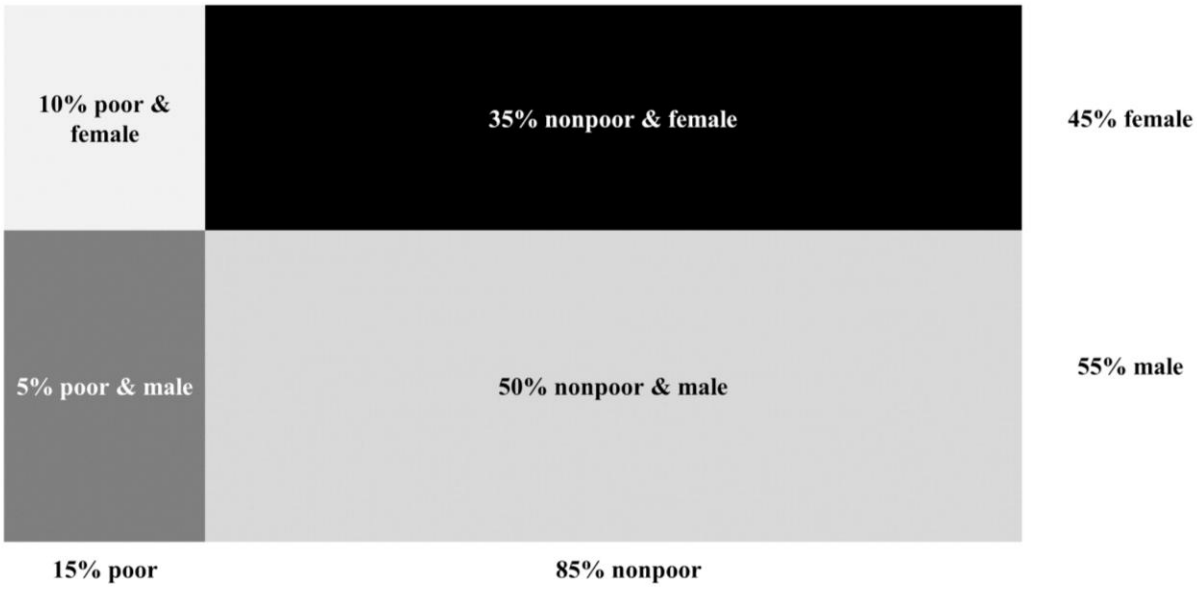


Figure 1A Distribution of poverty across gender.
Note: The gendered poverty ratio considering demographics equals (10/45)/(5/55) = 2.44

Table 2A Ranking of countries regarding iron deficiency for potato.

	Ranking from BPI	Ranking from IO
Malawi	1	9
Rwanda	2	1
Bolivia	3	2
Kyrgyzstan	4	4
Kazakhstan	5	6
Peru	6	3
Nepal	7	7
Kenya	8	10
Angola	9	36
Armenia	10	13
Tanzania	11	22
Azerbaijan	12	12
Lesotho	13	5
Democratic People's Republic of Korea	14	30
Tajikistan	15	19
Mongolia	16	17
Turkmenistan	17	33
Bangladesh	18	11
Uzbekistan	19	8
Algeria	20	16
Lebanon		14
India		15
Afghanistan		18
Colombia		20

Note: Iron deficiency was measured through the proportion of preschool-age children with haemoglobin less than 110g/dl and through Disability Adjusted Life Years (DALYs) per capita by Iron Deficiency Anaemia (IDA). Results for the BPI index were derived from **https://bpi.harvestplus.org/bpi_cropmaps.html?id=d11**. Results for the Impact Opportunities (IO) were derived from the Priority Index tab at **https://glomip.cgiar.org/impact-opportunities**. Date of extraction was 2024/01/09.

Appendix B – README for parser module

Raw data as downloaded from the original source is processed using the following steps:

A. transform tabular data to long format;
B. transform grid-level data to tabular long-format:
    - aggregate (either through summation or weighted averaging) grids to national-level;
    - crop-agnostic spatial data is first overlayed with the proportion of the area harvested per crop at grid-level (from Tang et al., 2024) to create national-crop-level relative breadth indicators;
C. remove irrelevant values:
    - for market segment-level data: entries with zero or missing area, yield or price, and entries for forages;
    - for all other data: duplicates and aggregates;
D. set consistent country and crop identifiers:
    - country names are defined using the shortest version and do not reflect official names;
    - cop names are taken from the Genesys Portal (Crop Trust, 2024);
E. standardise units[10]:
    - number of people (no thousands, millions, etc.);
    - percentages between 0-100%;
    - area in hectares (ha);
    - total environmental outputs in Mg (= metric tons);
    - relative environmental outputs in kg/ha;
    - agricultural outputs in mass units (kg) as farm-level;
F. balance data:
    - smooth (balance) national-crop data from FAO (except for indicators on NPK application) to remove outliers;
    - The values for yield and price refer either to dry or fresh product. See this overview for more details. Product refers to cereals, tubers, bananas or fodder (Forages (crop) Market Segments only).
    - set negative values to zero;
G. aggregate across time:
    - retain the most recent value per country or country-crop for every indicator (up to 2004);
    - take three-year average for national-crop data from FAO (except for indicators on NPK application);

[10] Nutrient units are measured as follows: energy in kilocalories (kcal), proteins in grams (g), zinc and iron in milligrams (mg) and vitamin A in micrograms (µg).

H. coalesce selected variables across different data sources (e.g. population or population undernourished) to maximise the number of observations;
I. calculate indicator if not directly available:
    - gender equity indicators, where possible, are calculated considering differences in gender demographics (see Figure 1A);
    - denominator for prevalence indicators always refer to national-level values;
J. transform indicators (if possible), except the cross-cutting ones so that larger values represent larger underdevelopment levels (same directionality);
K. impute missing values:
    - for national-level nonsummable indicators using the UN subregional average (scale-independent);
    - for national-level nonsummable indicators that are still missing using the UN regional average (scale-independent);
    - for national-level nonsummable indicators by multiplying the corresponding prevalence indicator with area or population;
    - for national-crop-level helper variables using the UN subregional, UN regional or World Bank regional average (whichever is needed) (scale-independent);
    - for national-crop-level relative breadth indicators by multiplying the corresponding prevalence indicator with area or population, correcting for differences in crop areas between FAOSTAT (2022) and Tang et al. (2024);
L. set consistent harvested area:
    - multiply relative breadth indicators by the area harvested;
    - for national-crop-level data, this is done to avoid differences in area harvested between data sources (reference: area harvested from FAO);
    - for market segment-level data, this is done to rescale breadth indicators from national-crop-level to market segment-level (reference: area harvested from expert elicitation);
M. remove countries not listed in FAO's Crop and Livestock Production database
N. adjust downscales at national-crop-level so that the national sum never exceeds 100%.
    - if the nationally summed downscales exceed 100% by X, then all downscales (for each crop within the country) get adjusted downward by X%.
O. Raster files are saved with standard settings:
    - extent: -180, 180, -90, 90
    - resolution: 0.05 arc degrees (3 arc-minutes) - same as CROPGRIDS
    - CRS: lon/lat WGS 84 (EPSG:4326)